

<https://github.com/Juan0987654321/Tasques-LMG/tree/main>

```
create database IF NOT EXISTS ex4t6;
```

```
use ex4t6;
```

```
CREATE TABLE accounts (  
    account_id INT PRIMARY KEY,  
    account_data JSON  
);
```

```
CREATE TABLE transactions (  
    txn_id INT PRIMARY KEY,  
    txn_data JSON  
);
```

```
INSERT INTO accounts VALUES
```

```
(1,{'avail_balance': 1057.75, "close_date": "NULL", "last_activity_date":  
"2005-01-04", "open_date": "2000-01-15", "pending_balance": 1057.75, "status":  
"ACTIVE", "cust_id": 1, "open_emp_id": 10, "open_branch_id": "2",  
"open_branch_name": "Woburn Branch", "open_branch_address": "422 Maple St.",  
"open_branch_city": "Woburn", "open_branch_state": "MA",  
"open_branch_zip_code": "01801", "product_cd": "CHK", "product_date_offered":  
"2000-01-01", "product_date_retired": "NULL", "product_name": "checking account",  
"product_type_cd": "ACCOUNT", "product_type_name": "Customer Accounts"}),
```

```
(2,{'avail_balance': 500, "close_date": "NULL", "last_activity_date": "2004-12-19",  
"open_date": "2000-01-15", "pending_balance": 500, "status": "ACTIVE", "cust_id": 1,  
"open_emp_id": 10, "open_branch_id": "2", "open_branch_name": "Woburn Branch",  
"open_branch_address": "422 Maple St.", "open_branch_city": "Woburn",  
"open_branch_state": "MA", "open_branch_zip_code": "01801", "product_cd": "SAV",  
"product_date_offered": "2000-01-01", "product_date_retired": "NULL",  
"product_name": "savings account", "product_type_cd": "ACCOUNT",  
"product_type_name": "Customer Accounts"}),
```

```
(3,{'avail_balance': 3000, "close_date": "NULL", "last_activity_date": "2004-06-30",  
"open_date": "2004-06-30", "pending_balance": 3000, "status": "ACTIVE", "cust_id":  
1, "open_emp_id": 10, "open_branch_id": "2", "open_branch_name": "Woburn  
Branch", "open_branch_address": "422 Maple St.", "open_branch_city": "Woburn",  
"open_branch_state": "MA", "open_branch_zip_code": "01801", "product_cd": "CD",  
"product_date_offered": "2000-01-01", "product_date_retired": "NULL",  
"product_name": "certificate of deposit", "product_type_cd": "ACCOUNT",  
"product_type_name": "Customer Accounts"}),
```

(4,{'avail_balance': 2258.02, "close_date": "NULL", "last_activity_date": "2004-12-27", "open_date": "2001-03-12", "pending_balance": 2258.02, "status": "ACTIVE", "cust_id": 2, "open_emp_id": 10, "open_branch_id": "2", "open_branch_name": "Woburn Branch", "open_branch_address": "422 Maple St.", "open_branch_city": "Woburn", "open_branch_state": "MA", "open_branch_zip_code": "01801", "product_cd": "CHK", "product_date_offered": "2000-01-01", "product_date_retired": "NULL", "product_name": "checking account", "product_type_cd": "ACCOUNT", "product_type_name": "Customer Accounts"}),

(5,{'avail_balance': 200, "close_date": "NULL", "last_activity_date": "2004-12-11", "open_date": "2001-03-12", "pending_balance": 200, "status": "ACTIVE", "cust_id": 2, "open_emp_id": 10, "open_branch_id": "2", "open_branch_name": "Woburn Branch", "open_branch_address": "422 Maple St.", "open_branch_city": "Woburn", "open_branch_state": "MA", "open_branch_zip_code": "01801", "product_cd": "SAV", "product_date_offered": "2000-01-01", "product_date_retired": "NULL", "product_name": "savings account", "product_type_cd": "ACCOUNT", "product_type_name": "Customer Accounts"}),

(7,{'avail_balance': 1057.75, "close_date": "NULL", "last_activity_date": "2004-11-30", "open_date": "2002-11-23", "pending_balance": 1057.75, "status": "ACTIVE", "cust_id": 3, "open_emp_id": 13, "open_branch_id": "3", "open_branch_name": "Quincy Branch", "open_branch_address": "125 Presidential Way", "open_branch_city": "Quincy", "open_branch_state": "MA", "open_branch_zip_code": "02169", "product_cd": "CHK", "product_date_offered": "2000-01-01", "product_date_retired": "NULL", "product_name": "checking account", "product_type_cd": "ACCOUNT", "product_type_name": "Customer Accounts"});

INSERT INTO transactions VALUES

(1,{'amount': 100, "funds_avail_date": "2000-01-15 00:00:00", "txn_date": "2000-01-15 00:00:00", "txn_type_cd": "CDT", "account_id": 1, "teller_emp_id": 1, "execution_branch_id": 1, "execution_branch_name": "Headquarters", "execution_branch_address": "3882 Main St.", "execution_branch_city": "Waltham", "execution_branch_state": "MA", "execution_branch_zip_code": "02451"}),

(2,{'amount': 500, "funds_avail_date": "2000-01-15 00:00:00", "txn_date": "2000-01-15 00:00:00", "txn_type_cd": "CDT", "account_id": 2, "teller_emp_id": 2, "execution_branch_id": 1, "execution_branch_name": "Headquarters", "execution_branch_address": "3882 Main St.", "execution_branch_city": "Waltham", "execution_branch_state": "MA", "execution_branch_zip_code": "02451"}),

(3,{'amount': 75, "funds_avail_date": "2004-06-30 00:00:00", "txn_date": "2004-06-30 00:00:00", "txn_type_cd": "CDT", "account_id": 3, "teller_emp_id": 2,

"execution_branch_id": 2, "execution_branch_name": "Woburn Branch",
"execution_branch_address": "422 Maple St.", "execution_branch_city": "Woburn",
"execution_branch_state": "MA", "execution_branch_zip_code": "01801"}),

(4,{ "amount": 1125, "funds_avail_date": "2001-03-12 00:00:00", "txn_date":
"2001-03-12 00:00:00", "txn_type_cd": "CDT", "account_id": 4, "teller_emp_id": 2,
"execution_branch_id": 1, "execution_branch_name": "Headquarters",
"execution_branch_address": "3882 Main St.", "execution_branch_city": "Waltham",
"execution_branch_state": "MA", "execution_branch_zip_code": "02451"}),

(5,{ "amount": 250, "funds_avail_date": "2001-03-12 00:00:00", "txn_date":
"2001-03-12 00:00:00", "txn_type_cd": "CDT", "account_id": 5, "teller_emp_id": 7,
"execution_branch_id": null, "execution_branch_name": null,
"execution_branch_address": null, "execution_branch_city": null,
"execution_branch_state": null, "execution_branch_zip_code": null}),

(6,{ "amount": 325, "funds_avail_date": "2002-11-23 00:00:00", "txn_date":
"2002-11-23 00:00:00", "txn_type_cd": "CDT", "account_id": 7, "teller_emp_id": 9,
"execution_branch_id": null, "execution_branch_name": null,
"execution_branch_address": null, "execution_branch_city": null,
"execution_branch_state": null, "execution_branch_zip_code": null}),

(7,{ "amount": 350, "funds_avail_date": "2002-12-15 00:00:00", "txn_date":
"2002-12-15 00:00:00", "txn_type_cd": "CDT", "account_id": 8, "teller_emp_id": 7,
"execution_branch_id": 4, "execution_branch_name": "So. NH Branch",
"execution_branch_address": "378 Maynard Ln.", "execution_branch_city": "Salem",
"execution_branch_state": "NH", "execution_branch_zip_code": "03079"}),

(8,{ "amount": 154, "funds_avail_date": "2003-09-12 00:00:00", "txn_date":
"2003-09-12 00:00:00", "txn_type_cd": "CDT", "account_id": 10, "teller_emp_id": 12,
"execution_branch_id": null, "execution_branch_name": null,
"execution_branch_address": null, "execution_branch_city": null,
"execution_branch_state": null, "execution_branch_zip_code": null}),

(9,{ "amount": 55, "funds_avail_date": "2000-01-15 00:00:00", "txn_date":
"2000-01-15 00:00:00", "txn_type_cd": "CDT", "account_id": 11, "teller_emp_id": 9,
"execution_branch_id": 4, "execution_branch_name": "So. NH Branch",
"execution_branch_address": "378 Maynard Ln.", "execution_branch_city": "Salem",
"execution_branch_state": "NH", "execution_branch_zip_code": "03079"}),

(10,{ "amount": 452, "funds_avail_date": "2004-09-30 00:00:00", "txn_date":
"2004-09-30 00:00:00", "txn_type_cd": "CDT", "account_id": 12, "teller_emp_id": 13,
"execution_branch_id": 2, "execution_branch_name": "Woburn Branch",

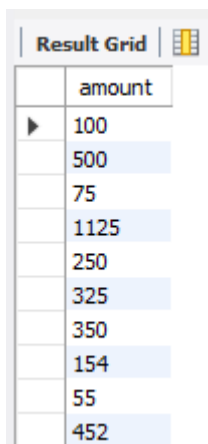
```
"execution_branch_address": "422 Maple St.", "execution_branch_city": "Woburn",  
"execution_branch_state": "MA", "execution_branch_zip_code": "01801"});
```

```
select * from ex4t6.transactions;
```

-- 2. Troba la consulta SQL utilitzant les funcions proporcionades per MySQL per a manejar codi JSON que retornaria la solució a les següents preguntes.

-- 2.1. Quina consulta utilitzaries per obtenir l'import (amount) de totes les transaccions de la taula transactions (Utilitza la funció JSON_UNQUOTE per a eliminar les cometes)?

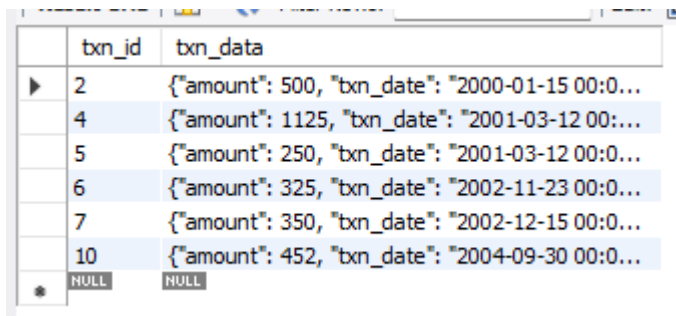
```
SELECT JSON_UNQUOTE(txn_data->'$.amount') AS amount FROM transactions;
```



	amount
▶	100
	500
	75
	1125
	250
	325
	350
	154
	55
	452

-- 2.2 Com podries filtrar les transaccions que tenen una quantitat superior a 200 (utilitza JSON_EXTRACT)?

```
SELECT * FROM transactions  
WHERE JSON_EXTRACT(txn_data, '$.amount') > 200;
```



	txn_id	txn_data
▶	2	{"amount": 500, "txn_date": "2000-01-15 00:0..."}
	4	{"amount": 1125, "txn_date": "2001-03-12 00:..."}
	5	{"amount": 250, "txn_date": "2001-03-12 00:0..."}
	6	{"amount": 325, "txn_date": "2002-11-23 00:0..."}
	7	{"amount": 350, "txn_date": "2002-12-15 00:0..."}
	10	{"amount": 452, "txn_date": "2004-09-30 00:0..."}
✱	HULL	HULL

-- 2.3 Escriu una consulta que obtingui l'ID de la transacció (txn_id) i la data de la transacció (txn_date) per totes les transaccions.

```
SELECT txn_id, JSON_UNQUOTE(txn_data->'$.txn_date') AS txn_date FROM  
transactions;
```

	txn_id	txn_date
▶	1	2000-01-15 00:00:00
	2	2000-01-15 00:00:00
	3	2004-06-30 00:00:00
	4	2001-03-12 00:00:00
	5	2001-03-12 00:00:00
	6	2002-11-23 00:00:00
	7	2002-12-15 00:00:00
	8	2003-09-12 00:00:00
	9	2000-01-15 00:00:00
	10	2004-09-30 00:00:00

-- 2.4 Escriu una consulta que recuperi l'ID del compte (account_id) i l'import (amount) de les transaccions, unint les taules transactions i accounts per account_id.

```
SELECT a.account_id, JSON_UNQUOTE(t.txn_data->'$.amount') AS amount
FROM accounts a JOIN transactions t ON
JSON_UNQUOTE(t.txn_data->'$.account_id') = a.account_id;
```

	account_id	amount
▶	1	100
	2	500
	3	75
	4	1125
	5	250
	7	325

-- 2.5 Com podries calcular l'import total de les transaccions per cada compte (account_id)?

```
SELECT
    JSON_UNQUOTE(txn_data->'$.account_id') AS account_id,

    SUM(JSON_UNQUOTE(txn_data->'$.amount')) AS total_amount
FROM transactions

GROUP BY account_id;
```

	account_id	total_amount
▶	1	100
	2	500
	3	75
	4	1125
	5	250
	7	325
	8	350
	10	154
	11	55
	12	452